

Azure vs. DigitalOcean

Choosing the Right Cloud Foundation for Essential Field Services

PeakLogic delivers intelligence, control, compliance automation, and operator-assistive workflows across distributed wastewater, environmental, and facility systems. The cloud foundation behind PeakLogic must support secure IoT ingestion, multi-tenant isolation, regulatory compliance, and SCADA-adjacent reliability. This brief summarizes why **Microsoft Azure** is the correct strategic platform compared to **DigitalOcean**.

Side-by-side comparison

CAPABILITY	AZURE	DIGITALOCEAN
IoT ingestion	✔ Azure IoT Hub (enterprise-grade)	✘ Not available
Device provisioning (DPS)	✔ Automated secure onboarding	✘ Not available
Device Update (ADU)	✔ Secure firmware lifecycle	✘ Not available
Enterprise identity (SSO/SCIM)	✔ Entra ID	✘ Not supported
Private networking	✔ VNETs, Private Endpoints	— Limited
Compliance certifications	✔ SOC 2, HIPAA, ISO, FedRAMP	— Minimal
Global redundancy	✔ 60+ regions	— Limited
Multi-tenant SaaS architecture	✔ First-class support	— DIY only
Event-driven telemetry pipelines	✔ Native services	✘ Not available
Cost	— Higher, enterprise-grade	— Lower, developer-friendly
Best for	— IoT platforms, enterprise SaaS, regulated industries	— Small apps, MVPs, simple hosting

Why Azure is the right direction for PeakLogic

- 1

Purpose-built for IoT at scale

Azure IoT Hub, DPS, and Device Update provide secure ingestion, provisioning, and lifecycle management for thousands of distributed devices. DigitalOcean has no IoT-native services.
- 2

Enterprise security & compliance

Entra ID, private networking, RBAC, and compliance certifications (SOC 2, HIPAA, ISO, FedRAMP) support regulated wastewater and municipal environments. DigitalOcean lacks enterprise identity and compliance depth.
- 3

Multi-tenant architecture for operators & municipalities

Azure PostgreSQL with Row-Level Security enables strict tenant isolation across operators, service providers, and municipalities. DigitalOcean requires custom engineering to approximate this.
- 4

SCADA-adjacent reliability

Azure provides global redundancy, SLAs, and event-driven pipelines required for critical operations. DigitalOcean is not designed for mission-critical industrial workloads.
- 5

Seamless integration with PeakLogic’s architecture

PeakLogic’s cloud-native design (IoT Hub → Functions → PostgreSQL → PeakView360) aligns directly with Azure’s strengths. DigitalOcean cannot support the ingestion, security, or compliance requirements of the platform.
- 6

Long-term scalability

Azure supports thousands of sites, millions of messages, and enterprise integrations. DigitalOcean is optimized for small apps and early-stage projects.

EXECUTIVE SUMMARY

Azure is a full-scale enterprise cloud platform built for IoT, security, compliance, and global scale.
DigitalOcean is a developer-friendly cloud built for simple applications.

For a platform delivering intelligence, control, compliance, and risk reduction across essential field services, Azure provides the reliability, security, and operational depth PeakLogic requires.